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Advanced Hemodynamic Modelling and Simulation of Flow Diverters in the Treatment of Intracranial Aneurysms

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an meine Familie und ganz besonders an meine Oma Hilde

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1.1 Conclusion

Decades of research in biomechanics have laid a foundation in clinical practice by developing virtual analysis tools for various cardiovascular diseases. These tools have not only enhanced our understanding of these pathologies but also facilitated the design of optimized treatment strategies. Building on this foundation we aim to improve diagnosis and treatment options for patients with intracranial aneurysms. This work has covered three research areas: (1) the study of stabilized finite elements for cardiovascular flows, (2) the assessment of the vascular extension around an aneurysm required for accurate numerical simulations, and (3) the development of a fast virtual stenting algorithm for deploying flow diverter stents. Although distinct, these aspects are united by a shared goal of advancing state-of-the-art modeling, with a strong emphasis on the numerical aspects of simulations.

After providing an overview on intracranial aneurysms in the introduction, we delve into the physics and numerical methods for simulating hemodynamics in brain arterial flows. In chapter 2 we revisited the existing numerical framework in a didactic manner, identifying an inconsistency in one modeling choice, which led to the research question: "How can a tensor-based stabilization parameter become impervious to the ordering of elements?" To address this, we developed a novel strategy based on Principal Component Analysis and tested it against common 2D and 3D examples from the literature. While not exhaustive, the validation showed improvements over an established alternative and consistently aligned well with experimental data.

With an improved understanding of the FEM backend, we explored the complex technical aspects of brain arterial flow, which is constrained by medical knowledge and limited measurement data. We reviewed the literature on boundary conditions and detailed the segmentation and meshing workflows implemented. This led to another research question: "What is the extent of the vasculature that needs to be simulated around the region of interest to obtain physiological results, particularly when using common outflow models?" This question arises from segmentating the interconnected vessel network in the brain, where arterial loops can theoretically direct flow in any direction and do not always follow common splitting laws. We studied the sensitivity of three patient-specific cases with aneurysms, comparing full Circle of Willis (CoW) simulations with two reduced vasculature models commonly seen in the literature. The results revealed case-dependent variability, suggesting to check MR images to assess the completeness of the CoW and in cause of doubt simulate the network to obtain estimates of the flow direction and magnitude.

The next focus was on flow diverter stents, which enable clinicians to treat otherwise inoperable intracranial aneurysms. A numerical representation of these devices not only replicates altered flow conditions but also provides insights into how hemodynamic changes affect vessel walls. We proposed a novel virtual stenting algorithm based on geometric analysis and validated it against literature cases, in-vitro experiments, and postoperative patient images. We then demonstrated how to embed the device into an arbitrary unstructured computational mesh and ran a test simulation. The proposed methodology offers three advantages: (i) accurate estimation of local stent porosities, (ii) robust and fast deployment, and (iii) reliance only on centerline information and manufacturer data. However, the method does not account for mechanical interactions with vessel walls and may underperform with significant stent undersizing.

Equipped with the essential tools for post-treatment aneurysm simulations, we examined three patient cases involving flow diverter stents with varying outcomes. The first case, featuring a giant aneurysm, suffered a delayed rupture after treatment. Simulations of preoperative and postoperative states including wall compliance were carried out and served an analysis based on several rupture hypotheses from the literature. The remaining cases involved (i) a ruptured blister aneurysm and (ii) a ruptured wide-necked aneurysm. Although different, both cases required assessing similar risks, including the trade-offs between reducing flow stress and managing thromboembolic complications. We report on the computed hemodynamics and include a study on deploying a second stent over the existing one. With these results, we hope to contribute to larger population studies aimed at optimizing treatment options.

1.2 Perspectives

As one of the pioneering theses in bio-fluid mechanics at our research lab, this work has met the envisioned goal but also left several promising avenues unexplored due to time constraints. The most compelling opportunities for further research lie in modeling and optimization.

Modelling

Thrombosis plays a crucial role in the treatment of intracranial aneurysms, influencing outcomes such as occlusion, autolysis, and thromboembolic complications. Simulating thrombosis in stented aneurysms could provide vital information on the timescale of autolysis and its potential impact on the aneurysm. Additionally, estimating the risk of thromboembolism following flow diverter deployment is essential. Here, simulations could offer statistical insights into how uncovered metallic areas increase the likelihood of complications. Although thrombosis simulations are well-established in the literature [1, 2], integrating these time-intensive processes with large simulation meshes, such as those post-stenting, remains a challenge. Continuum models of the stent based on local porosity maps [3] offer a promising approach to this challenge and could complement the tools developed in this work.

Another critical area for future research is the improvement of boundary conditions and segmentation techniques, both of which have been recognized by the community as significant factors in the accuracy of simulations. We have made efforts to incorporate multiple sources of information to enhance the accuracy of these aspects, particularly in segmentation. Looking ahead, advancements in imaging technology and the availability of patient-specific boundary conditions could further increase the reliability of simulation results, thereby strengthening the conclusions drawn from them.

Optimization

Optimizing treatment is a key objective in the development of medical devices. Although not discussed in detail in this thesis, we have initiated a pipeline to enhance the design of flow diverters using Reinforcement Learning algorithms [4]. This preliminary work focused on adjusting strut positioning to achieve targeted reductions in wall shear stress within aneurysms. Since then, we have refined the computational framework and advanced the Machine Learning components of the optimization process [5] thanks to the funding from the MSCA AI4TheSciences programme at PSL University. Building on these advancements and incorporating additional targets, such as thrombosis risk estimations, could lead to the development of optimized, patient-specific devices.

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RÉSUMÉ

Les anévrismes intracrâniens sont courants mais mal compris, malgré les graves conséquences d'une rupture. La complexité des processus biochimiques du cerveau et la délicatesse de son environnement limitent les études in vivo. La dynamique des fluides numérique (CFD) peut aider à surmonter ces défis en reproduisant les conditions d'écoulement dans le cerveau, qui sont censées influencer l'initiation, la croissance et la rupture des anévrismes. Les simulations CFD peuvent également tester des traitements hypothétiques et évaluer leur impact. Cette thèse se concentre sur le développement d'un cadre de simulation de haute fidélité pour modéliser l'hémodynamique cérébrale, dans le but d'améliorer notre compréhension des anévrismes intracrâniens et de perfectionner leur traitement.

La thèse commence par explorer les aspects médicaux des anévrismes cérébraux, y compris leurs différents types, options de traitement et risques associés. Elle aborde ensuite la physique et les méthodes numériques qui constituent la base du cadre de simulation, en proposant de nouvelles contributions méthodologiques. Avec ces connaissances et validations en main, le travail progresse vers l'adaptation des simulations à l'environnement vasculaire du cerveau, en discutant des aspects critiques tels que la segmentation, les conditions aux limites et l'étendue du domaine simulé. Pour recréer virtuellement le déploiement d'un stent de dérivation, un algorithme basé sur des contraintes géométriques est développé et rigoureusement validé à l'aide d'exemples tirés de la littérature, de données expérimentales et de cas réels de patients. Dans le dernier chapitre, ces outils sont appliqués pour calculer l'hémodynamique dans trois cas complexes de patients présentant des anévrismes rompus. Les effets du traitement par stent de dérivation sont analysés et comparés aux données médicales les plus récentes.

MOTS CLÉS

Anévrismes intracrâniens, hémodynamique, dynamique des fluides numérique, méthode des éléments finis, méthode variationnelle multi-échelle

ABSTRACT

Intracranial aneurysms are common but not well understood, despite the severe consequences of rupture. The brain's complex biochemical processes and delicate environment limit in-vivo studies. Computational Fluid Dynamics (CFD) can help overcome these challenges by replicating the brain's flow conditions, which are thought to influence the initiation, growth, and rupture of aneurysms. CFD simulations can also test hypothetical treatments and evaluate their impact. This thesis focuses on developing a high-fidelity computational framework to simulate brain hemodynamics, aiming to enhance our understanding of intracranial aneurysms and improve their treatment.

The thesis begins by exploring the medical aspects of cerebral aneurysms, including their various types, treatment options, and associated risks. It then delves into the physics and numerical methods that form the foundation of the simulation framework, offering new contributions to the methodology. With this knowledge and validation in hand, the work progresses to adapt the simulations to the brain's vascular environment, discussing critical aspects such as segmentation, boundary conditions, and the extent of the simulated domain. To virtually recreate the deployment of a flow diverter stent, an algorithm based on geometrical constraints is developed and thoroughly validated using literature examples, experimental data, and real patient cases. In the final chapter, these tools are applied to compute the hemodynamics in three complex patient cases involving ruptured aneurysms. The effects of flow diverter treatment are analyzed and compared with state-of-the-art medical literature.

KEYWORDS

Intracranial aneurysms, Haemodynamics, Computational Flud Dynamics, Finite Elements, Variational Multiscale Method